The long-root multiplier is a coboundary,

and therefore the long-root poles of the structure constants are *removable*:
a retraction of my own pre-registered answer, and a confirmation of Rick's hunch

with the exact family of renormalisations that remove them — $\chi(x)\chi(1/x) = \rho^{-1}$ *and* $\chi(\varepsilon) = 1$ for $\varepsilon^h = 1$

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17 October 2026

PROTOCOL §2.3 record.

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This PDF corresponds to the pushed commit above. Where this text and the commit disagree, **the commit is authoritative**.

Abstract

On 2026-10-17 Rick (UID 819) proved **Lemma QC**: on the long-root divisor $H_{ab} = \{w_a w_b = 1\}$, Motegi's normalised type-C weight functions satisfy $\widetilde{A}_{\mathbf{M},\mathbf{I}} = g \cdot \widetilde{A}_{\mathbf{M}\setminus\{a,b\},\mathbf{I}}$ for *every* $\mathbf{I}$, with $g = (1-q)^2/((1-qw_a^2)(1-qw_a^{-2}))$. This is my ρ^C, reached by a different route, and it upgrades my Lemma A from a divisibility to an identity. He attached a hunch: *"Clio's poles measure the normalisation defect; a geometric normalisation would have no long-root poles at all."*

I pre-registered the answer **no**, with a mechanism. **The pre-registered answer is wrong and is retracted here.** Rick is right, and the reason is a one-line observation about ρ that I had already proved and had not read correctly: writing $\varphi(v) := (1-q)/(1-qv^h)$,

$$\rho^X = \varphi(w_a)\,\varphi(w_b),$$

a product of a *single universal one-variable function* evaluated at the two colliding indices. That is exactly the statement that the collapse multiplier is a **coboundary**. Setting $\psi := \varphi^{-1}$ and $\Psi(\mathbf{M}) := \prod_{m\in\mathbf{M}} \psi(w_m)$, the renormalised basis $\widehat{A}_{\mathbf{M},\mathbf{I}} := \widetilde{A}_{\mathbf{M},\mathbf{I}}\Psi(\mathbf{M})$ collapses with multiplier 1 for every long-root pair simultaneously (Theorem 2), *and* every one of its structure constants is regular on every long-root divisor (Theorem 7). Measured at $N = 3$, type C: the baseline census returns 13 non-Laurent constants out of 115, with poles on long-root divisors only; after the renormalisation the count is **0/115**.

The exact family is determined (Theorem 13): among renormalisations $\Psi_\chi(\mathbf{M}) = \prod_{m\in\mathbf{M}} \chi(w_m)$, the long-root poles vanish *iff* $\chi(x)\chi(1/x) = \rho^{-1}|_H$, and — a condition the session did not anticipate and the data forced — the *short*-root divisors $\{w_m = \varepsilon\}$, $\varepsilon^h = 1$, stay pole-free *iff* $\chi(\varepsilon) = 1$. The short-root collapse multiplier is measured to be exactly 1. Both conditions are met by ψ for the same reason: $\psi(\varepsilon) = 1$ precisely when $\varepsilon^h = 1$, which is the locus $\{\rho = 1\}$.

Consequences. (i) Q491's criterion "$c^\nu_{\lambda\mu}$ Laurent $\iff$ $|\nu| - \max(|\lambda|, |\mu|) \le 1$" is a property of the $\mathcal{E}^C$-normalised basis, **not** of the algebra: under ψ it becomes vacuous. The negative

half of Q500 — a "Pfaffian-category obstruction" read off the non-Laurentness — is withdrawn in the form in which I stated it. (ii) What survives, and is the right invariant, is *not* the pole but the *class*: the poles are removable because ρ is a coboundary, and had ρ not split as $\varphi(w_a)\varphi(w_b)$ no point-diagonal renormalisation could have removed them. The obstruction I had been measuring was a cochain. The class is zero.

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1 What this session set out to test, and what happened

The brief (`state/PROVE.md`, 2026-10-17 c2) set three targets. Theorem A, expected easy: ρ^X is a coboundary. Theorem B, *the actual question*: does the same renormalisation remove the poles of the *structure constants*? Theorem C, conditional on B: is the pole order invariant across the whole family of diagonal renormalisations?

The brief pre-registered the answer to B as **no**, with this mechanism, quoted verbatim:

> "the renormalisation is a diagonal rescaling of the basis by $\prod_{m\in\mathbf{M}}\psi(w_m)$, so a structure constant transforms by a ratio of ψ-products. $\psi(w) = (1-qw^h)/(1-q)$ has … no zeros or poles on the collision divisor $w_a w_b = 1$ at generic w_a … A factor that is regular and non-vanishing at the generic point of H_{ab} **cannot cancel a pole there**."

The final sentence is true. The sentence before it is false, and §10 isolates the error exactly: the product is indexed by $m \in \mathbf{M}$ — the *specialisation set*, i.e. the *point* — and a rescaling indexed by the point is not a rescaling of the basis. A rescaling of the basis is indexed by $\mathbf{I}$, the *label*. The two operations do different things to structure constants, and, decisively, **Theorem A is true only for the point-indexed reading while the Theorem B argument is valid only for the label-indexed one**. One symbol, two readings, two paragraphs apart, each locally plausible.

So Theorem B is false as pre-registered, and its negation is a theorem. Theorem C is then settled in the opposite direction to the brief's guess: the pole order is *not* invariant across the family — it is 1 off a distinguished coset of renormalisations and 0 on it — and the coset is computed exactly.

2 Task 0: which object is Rick's $\widetilde{A}$?

The brief imposed a hard 30-minute gate here, because the 2026-10-17 c1 retraction was caused by exactly this class of error (a type-B weight function divided by the type-C normaliser), and because Rick's g matching my ρ^C has the shape of a one-coordinate match. I read `peers/rick/proofs/2026-10-17-q488-c` first-hand. All three answers are "same object".

1. **Normaliser.** His fact (M2), §1: "`E^C` (L1057–1062); $\widetilde{A} = WC/E^C$." Type C weight function divided by the type C normaliser. Mine likewise. (This is the coordinate on which c1 failed, and it is clean here.)

2. **Index convention.** His §1 defines $\phi_{\mathbf{I}}(\mathbf{M}) := \widetilde{A}_{\mathbf{M},\mathbf{I}} := \widetilde{WC}_{N,n,|\mathbf{I}|}(w_{\mathbf{M}} \mid w \mid \mathbf{I})$. The subscript $\mathbf{M}$ enters as the *specialisation* $u = w_{\mathbf{M}}$ and $\mathbf{I}$ as the *label* carried by W. That is my convention; his UID 819 Ask (1) asked me to check precisely this and the answer is that his $\mathbf{I}$ is my $\mathbf{I}$.

3. **Divisor.** His $H_{ab} = \{w_a w_b = 1\}$, local parameter $1 - 1/(w_a w_b)$; my H is the same divisor with the same parameter.

I also checked the arithmetic rather than only the names: with $h = 2$, $\rho^C = (1-q)^2/((1-qw_a^2)(1-qw_b^2))$ restricted to $w_b = w_a^{-1}$ is $(1-q)^2/((1-qw_a^2)(1-qw_a^{-2})) = g$. Identical.

Remark 1 (grading). Lemma QC is graded **proved** and not `peer-claimed`, because I read his proof (§3 of the proof file) rather than its statement. His §3 is a term-by-term comparison of the symmetrisation formula at $n = s$ and $n = s-2$: the head-pair factor $(u_j u_k - 1)$ kills every permutation putting both of $x, 1/x$ in the head, the surviving head selections are exactly those of the $(s-2)$-formula, and the leftover bracket is independent of the head selection. I have not re-derived it line by line; I have checked its two consumable outputs (the multiplier, and uniformity in $\mathbf{I}$) against my own engine, which reaches the same ρ^C by a different route. Independent corroboration in the strong sense: different author, different derivation.

3 Setup

Fix $N \geq 1$. Let $F := \mathbb{C}(q, w_1, \ldots, w_N)$, let $\mathcal{P} := 2^{[N]}$ be the set of subsets of $[N]$, and let

$$\mathcal{A} := F^{\mathcal{P}}$$

be the algebra of F-valued functions on $\mathcal{P}$ under pointwise operations — the "GKM algebra" of the 2^N fixed points. For $\mathbf{I} \subseteq [N]$ put $\phi_{\mathbf{I}} \in \mathcal{A}$, $\phi_{\mathbf{I}}(\mathbf{M}) := \widetilde{A}_{\mathbf{M},\mathbf{I}}$, Motegi's normalised weight function ($\widetilde{A}^X := W^X/\mathcal{E}^X$, evaluated at $u = w_{\mathbf{M}}$; n-independent by stability, Motegi L2823–2848). Two facts from Motegi, arXiv:2610.15344, both deep-read:

(M3) Triangularity (Thm 2.6(i), L603–620). $\widetilde{A}_{\mathbf{M},\mathbf{I}} = 0$ unless $\mathbf{M} \geq \mathbf{I}$ in Motegi's order ($|\mathbf{M}| \geq |\mathbf{I}|$ and the top $|\mathbf{I}|$ elements of $\mathbf{M}$ dominate $\mathbf{I}$ entrywise).

(M4) The diagonal (eq. `normalizationWC`, L1078). $\widetilde{A}_{\mathbf{I},\mathbf{I}}$ is a product over the positive roots of C_N. Its factors are $\prod_{j \leq k}(1 - w_{i_j}^{-1} w_{i_k}^{-1})$ over $\mathbf{I}$ — note $j \leq k$, so the *diagonal* term $(1 - w_i^{-2})$ occurs — together with A-type factors $(1 - w_j/w_{i_l})$ and q-factors $(1 - q \cdot \text{monomial})$.

By (M3)+(M4), $\{\phi_{\mathbf{I}}\}_{\mathbf{I}\subseteq[N]}$ is an F-basis of $\mathcal{A}$, and the *structure constants* $c^{\nu}_{\lambda\mu} \in F$ are defined by $\phi_\lambda \phi_\mu = \sum_\nu c^{\nu}_{\lambda\mu}\phi_\nu$.

Two families of divisors in w-space will matter. For $a < b$ the **long-root divisor** $H_{ab} := \{w_a w_b = 1\}$, with local ring $\mathcal{O}_{ab} \subset F$, uniformiser $t_{ab} := 1 - (w_a w_b)^{-1}$, residue field k_{ab}; for $\mathbf{M} \supseteq \{a, b\}$ write $\mathbf{M}'' := \mathbf{M} \setminus \{a, b\}$. For $m \in [N]$ and $\varepsilon^h = 1$ the **short-root divisor** $S^{\varepsilon}_m := \{w_m = \varepsilon\}$; for $\mathbf{M} \ni m$ write $\mathbf{M}^- := \mathbf{M} \setminus \{m\}$. On H_{ab} we abbreviate $x := w_a$, so $w_b = x^{-1}$.

(QC) Long-root collapse. For all $a < b$, all $\mathbf{M} \supseteq \{a, b\}$, all $\mathbf{I}$: $\phi_{\mathbf{I}}(\mathbf{M}) \equiv \rho^X \, \phi_{\mathbf{I}}(\mathbf{M}'')$ (mod $w_a w_b - 1$), where

$$\rho^X = \frac{(1-q)^2}{(1-qw_a^h)(1-qw_b^h)}, \qquad h = 2 \text{ (type } C), \quad h = 1 \text{ (type } B).$$

Type C is Rick's Lemma QC (§2) and my `2026-10-16-c2-longroot-multiplier.tex` Thm `collapse`; type B is `2026-10-17-typeB-multiplier-and-boundary-entry.tex` Thm `rhoB`. Write $g := \rho^X|_{H_{ab}} = (1-q)^2/((1-qx^h)(1-qx^{-h})) \in k_{ab}^\times$.

4 Theorem A: the multiplier is a coboundary

Theorem 2 (ρ splits)**.** *Put* $\varphi(v) := \dfrac{1-q}{1-qv^h} \in \mathbb{C}(q)(v)$ *and* $\psi := \varphi^{-1}$*, i.e.* $\psi(v) = \dfrac{1-qv^h}{1-q}$*. Then*

$$\rho^X = \varphi(w_a)\,\varphi(w_b) \qquad \text{for every pair } a < b,$$

with the same *one-variable* φ *for every pair. Consequently, setting* $\Psi(\mathbf{M}) := \prod_{m\in\mathbf{M}} \psi(w_m)$ *and* $\widehat{A}_{\mathbf{M},\mathbf{I}} := \widetilde{A}_{\mathbf{M},\mathbf{I}}\,\Psi(\mathbf{M})$*, we have for every long-root pair* $a < b$ *simultaneously and every* $\mathbf{I}$

$$\widehat{A}_{\mathbf{M},\mathbf{I}}\big|_{H_{ab}} = \widehat{A}_{\mathbf{M}'',\mathbf{I}}\big|_{H_{ab}} :$$

honest, untwisted GKM collapse, multiplier exactly 1*.*

Proof. $\varphi(w_a)\varphi(w_b) = (1-q)^2/((1-qw_a^h)(1-qw_b^h)) = \rho^X$ by inspection. For the consequence, $\Psi(\mathbf{M}) = \Psi(\mathbf{M}'')\psi(w_a)\psi(w_b)$ since $\mathbf{M} = \mathbf{M}'' \sqcup \{a, b\}$, so by (QC)

$$\widehat{A}_{\mathbf{M},\mathbf{I}}\big|_H = \psi(w_a)\psi(w_b)\,\Psi(\mathbf{M}'')\cdot\rho^X\,\widetilde{A}_{\mathbf{M}'',\mathbf{I}}\big|_H = \big(\varphi(w_a)\varphi(w_b)\big)^{-1}\rho^X\cdot\widehat{A}_{\mathbf{M}'',\mathbf{I}}\big|_H = \widehat{A}_{\mathbf{M}'',\mathbf{I}}\big|_H. \qquad \square$$

Remark 3 (what the content is)*.* The algebra is one line; the content is the *quantifier*. A priori the collapse multiplier on H_{ab} is a function attached to the *pair* $\{a, b\}$. Theorem 2 says it is the product, over the two colliding indices, of a single universal function of one variable. In the language of §5: the 1-cochain $\{(\mathbf{M}, ab) \mapsto \rho\}$ on the collision groupoid is the coboundary of the 0-cochain Ψ^{-1}. Had ρ depended on w_c for some $c \notin \{a, b\}$, or failed to factor by variable, *no* renormalisation of this shape could trivialise it, and everything below would fail. Note also $\psi|_{q=0} = 1$, so the renormalisation is invisible at $q = 0$: it is a purely q-theoretic twist, consistent with Rick's observation that "at $q = 0$ the twist is 1 (GKM)."

5 Collision weight, and the lattice lemma

Definition 4. Fix $a < b$ and $d \in \mathbb{Z}$. A function $f \in \mathcal{O}^{\mathcal{P}}_{ab}$ has **collision weight** d **at** H_{ab} if $f(\mathbf{M})|_H = g^d f(\mathbf{M}'')|_H$ for every $\mathbf{M} \supseteq \{a, b\}$. Write $W_d(ab)$ for the set of such f.

Immediately: each $W_d(ab)$ is an $\mathcal{O}_{ab}$-module (the scalar does not depend on $\mathbf{M}$); $W_d \cdot W_e \subseteq W_{d+e}$ under pointwise product; and W_0 is a subring containing 1. By (QC), $\phi_{\mathbf{I}} \in W_1(ab)$ for every $\mathbf{I}$; note in particular $1 \notin W_1$ unless $g = 1$, so W_1 is *not* a ring — and that failure is the whole phenomenon.

Proposition 5 (lattice lemma; Rick, Prop 5, re-derived)**.** $\mathrm{span}_{\mathcal{O}_{ab}}\{\phi_{\mathbf{I}} : \mathbf{I} \subseteq [N]\} = W_1(ab)$. *Consequently, for $f \in \mathcal{O}_{ab}^{\mathcal{P}}$ with coordinates $\ell_\nu(f)$ in the basis $\{\phi_{\mathbf{I}}\}$:*

$$\ell_\nu(f) \in \mathcal{O}_{ab} \text{ for all } \nu \iff f \in W_1(ab),$$

and any pole that does occur is simple.

Proof. "$\subseteq$": the $\phi_{\mathbf{I}}$ are Laurent polynomials in w over $\mathbb{C}[q]$ localised at (q), hence lie in $\mathcal{O}_{ab}^{\mathcal{P}}$, and satisfy the congruence by (QC); W_1 is an $\mathcal{O}_{ab}$-module.

Colength of W_1. The *defect map* $D : \mathcal{O}_{ab}^{\mathcal{P}} \to k_{ab}^{\{\mathbf{M} \supseteq \{a,b\}\}}$, $D_{\mathbf{M}}(f) := (f(\mathbf{M}) - g\, f(\mathbf{M}''))|_H$, is $\mathcal{O}_{ab}$-linear with kernel W_1 by definition. It is surjective: for $\mathbf{M}_0 \supseteq \{a,b\}$ take f the indicator of the single point $\mathbf{M}_0$; then $D_{\mathbf{M}}(f) = \delta_{\mathbf{M},\mathbf{M}_0}$, because $\mathbf{M}'' = \mathbf{M}_0$ would force $\mathbf{M} = \mathbf{M}_0 \cup \{a,b\} = \mathbf{M}_0$, excluded. Hence $\mathrm{length}(\mathcal{O}_{ab}^{\mathcal{P}}/W_1) = \#\{\mathbf{M} \supseteq \{a,b\}\} = 2^{N-2}$.

Colength of the span. Over the DVR $\mathcal{O}_{ab}$ the colength of the lattice spanned by the columns of $(\phi_{\mathbf{I}}(\mathbf{M}))_{\mathbf{M},\mathbf{I}}$ is $v_{ab}(\det)$. By (M3) the matrix is triangular for Motegi's order, so $\det = \pm \prod_{\mathbf{I}} \widetilde{A}_{\mathbf{I},\mathbf{I}}$, and by (M4) $v_{ab}(\widetilde{A}_{\mathbf{I},\mathbf{I}}) = 1$ if $a, b \in \mathbf{I}$ (the single factor $(1 - w_a^{-1} w_b^{-1})$) and 0 otherwise: at generic q no other factor of the positive-root product vanishes identically on H_{ab} — in particular $(1 - q w_a w_b)|_H = 1 - q \neq 0$ and $(w_a - w_b)|_H = w_a - w_a^{-1} \not\equiv 0$. Hence $v_{ab}(\det) = \#\{\mathbf{I} \supseteq \{a,b\}\} = 2^{N-2}$.

Two $\mathcal{O}_{ab}$-submodules of $\mathcal{O}_{ab}^{\mathcal{P}}$, one contained in the other, with equal finite colength, are equal. The displayed equivalence follows: if all $\ell_\nu(f)$ are regular then $f = \sum \ell_\nu(f) \phi_\nu$ lies in the span; conversely the span's elements have regular coordinates by uniqueness. Simplicity of the poles: $t_{ab} \cdot \mathcal{O}_{ab}^{\mathcal{P}} \subseteq W_1$. □

Remark 6 (the two numerical inputs, checked independently). Both inputs to the colength count were verified directly against the engine's printed diagonals at $N = 3$, type C (file `code-1016-q492/q492_N3_census`). For H_{12} the factor $(w_1 w_2 - 1)$ occurs in exactly $\widetilde{A}_{\mathbf{I},\mathbf{I}}$ for $\mathbf{I} = (1,2)$ and $\mathbf{I} = (1,2,3)$, each once: $2 = 2^{3-2}$. Likewise for H_{13} and H_{23}.

6 Theorem B: the long-root poles are removable

Theorem 7 (Rick's hunch, confirmed for long roots)**.** *Let $\chi \in \mathbb{C}(q)(v)^\times$ satisfy*

$$\textbf{\textit{(COB)}} \qquad \chi(x)\,\chi(1/x) = g(x)^{-1} \quad \textit{identically in } x,$$

let $\Psi_\chi(\mathbf{M}) := \prod_{m \in \mathbf{M}} \chi(w_m)$ and $\widehat{\phi}_{\mathbf{I}} := \Psi_\chi \phi_{\mathbf{I}}$, and assume $\Psi_\chi \in \mathcal{O}_{ab}^{\mathcal{P}}$. Let $\widehat{c}_{\lambda\mu}^{\,\nu}$ be the structure constants of $\{\widehat{\phi}_{\mathbf{I}}\}$. Then for every λ, μ, ν and every $a < b$, $\widehat{c}_{\lambda\mu}^{\,\nu}$ is regular at H_{ab}.

In particular $\chi = \psi$ satisfies (COB) *by Theorem 2, for every pair $a < b$ at once.*

Proof. $\Psi := \Psi_\chi$ is invertible in $\mathcal{A}$ (each $\Psi(\mathbf{M}) \in F^\times$), so

$$\widehat{\phi}_\lambda \widehat{\phi}_\mu = \sum_\nu \widehat{c}^{\,\nu} \widehat{\phi}_\nu \iff \Psi^2 \phi_\lambda \phi_\mu = \Psi \sum_\nu \widehat{c}^{\,\nu} \phi_\nu \iff \Psi \cdot \phi_\lambda \phi_\mu = \sum_\nu \widehat{c}^{\,\nu} \phi_\nu,$$

i.e. $\widehat{c}^{\,\nu} = \ell_\nu(f)$ with $f := \Psi \cdot \phi_\lambda \phi_\mu$. Now compute the weight of f at H_{ab}. Since $\mathbf{M} = \mathbf{M}'' \sqcup \{a,b\}$ and $w_b = x^{-1}$ on H,

$$\Psi(\mathbf{M})\big|_H = \chi(x)\chi(1/x)\,\Psi(\mathbf{M}'')\big|_H = g^{-1}\Psi(\mathbf{M}'')\big|_H,$$

so $\Psi \in W_{-1}(ab)$; and $\phi_\lambda \phi_\mu \in W_1 \cdot W_1 = W_2(ab)$ by (QC). Hence $f \in W_{-1} \cdot W_2 = W_1(ab)$. By Proposition 5, every $\ell_\nu(f)$ is regular at H_{ab}. □

Remark 8 (why this is not magic). The mechanism is entirely visible in Rick's Theorem 6. A product of two basis elements collides with multiplier g^2 while the span collides with g; the mismatch $g^2 - g = g(g-1)$ *is* the pole. Multiplying everything by a function of weight -1 moves the product to weight 1 and the span to weight 0 — no, more precisely: it moves the product $\Psi\phi_\lambda\phi_\mu$ to weight 1, which is the weight of the span, and the mismatch disappears. Equivalently: $\{\widehat{\phi}_{\mathbf{I}}\}$ spans a *ring* of weight-0 functions, and a ring has regular structure constants.

7 The short-root condition, which the data forced

Theorem 7 says nothing about the other divisors of $\det \widetilde{A}$. I did not expect that to matter, and it does. The $\chi = v\,\psi(v)$ arm — chosen because $\eta(v) = v$ satisfies $\eta(x)\eta(1/x) = 1$ and therefore preserves (COB) — removed *every* long-root pole as predicted and produced 57 new ones, all on divisors $\{w_i = -1\}$.

Theorem 9 ((SRC) short-root collapse, type C)**.** *For every N, every $\mathbf{I}$, every $\mathbf{M} \ni m$ and every ε with $\varepsilon^2 = 1$,*

$$\widetilde{A}_{\mathbf{M},\mathbf{I}}\big|_{w_m=\varepsilon} = \widetilde{A}_{\mathbf{M}^-,\mathbf{I}}\big|_{w_m=\varepsilon}\,:$$

the short-root collision multiplier is exactly 1*, uniformly in* $\mathbf{M}$ *and* $\mathbf{I}$*.*

Proof. Write $s := |\mathbf{M}|$, $r := |\mathbf{I}|$, $R := \mathbf{M}^-$, and run Motegi's symmetrisation formula unpadded at $n = s$ and at $n = s-1$ (stability, L2823–2848), with $\beta = 1$, $\gamma = 0$. Throughout, $\varepsilon^{-1} = \varepsilon$.

Step 1: ε must be a tail. The head factor of line 1 is $(1-q)(u_j^2 - 1)$, which vanishes at $u_j = \varepsilon$. So every permutation placing ε in a head position contributes 0. The $n - r$ tail positions are symmetric, so the $n - r$ ways of seating ε in a tail give equal contributions and cancel the change $1/(n-r)! \to 1/(n-1-r)!$. The surviving head selections are exactly the head selections of the $(s-1)$-formula. (If $s - 1 < r$ there is no tail slot at all, every permutation dies, and both sides are 0.)

Step 2: the extra factors are head-independent. Fix a head selection and compare the s-term with the $(s-1)$-term. The w-alphabet is unchanged, so line 1 matches. The extra factors carried by the tail ε are

$$\underbrace{\textstyle\prod_{k=1}^{N}(\varepsilon - qw_k)}_{\text{line 2b}} \cdot \underbrace{\textstyle\prod_{\text{heads } h} \frac{q\varepsilon - h}{\varepsilon - h}(\varepsilon h - 1)}_{\text{line 3a}} \cdot \underbrace{\textstyle\prod_{\text{tails } t \neq \varepsilon}(\varepsilon t - q)}_{\text{line 3b}} \cdot \underbrace{\varepsilon}_{\text{line 3c}}\,.$$

Now $\varepsilon h - 1 = \varepsilon(h - \varepsilon) = -\varepsilon(\varepsilon - h)$, so the line-3a factor collapses to $(-\varepsilon)^r \prod_h (q\varepsilon - h) = \varepsilon^r \prod_h (h - q\varepsilon)$; and $\varepsilon t - q = \varepsilon(t - q\varepsilon)$, so line 3b is $\varepsilon^{\,n-r-1} \prod_{t \neq \varepsilon}(t - q\varepsilon)$. The head product and the tail product therefore **recombine into a product over all of** R:

$$\varepsilon^r \cdot \varepsilon^{\,n-r-1} \prod_{v \in R}(v - q\varepsilon) = \varepsilon^{\,s-1} \prod_{v \in R}(v - q\varepsilon),$$

with no trace of the head selection. This is the exact analogue of step 2 of Rick's Lemma QC, and it is the only step that could have failed.

Step 3: the ratio. The prefactor $\prod_k u_k^{-(N+n)}$ contributes $\varepsilon^{-(N+s)} \prod_{v \in R} v^{-1}$. Collecting, and using $v^{-1}(v - q\varepsilon) = 1 - q\varepsilon/v$ and $\varepsilon^{-N} \prod_k (\varepsilon - qw_k) = \prod_k (1 - q\varepsilon w_k)$,

$$\left.\frac{F_s}{F_{s-1}}\right|_{w_m=\varepsilon} = \prod_{k=1}^{N}(1 - q\varepsilon w_k) \prod_{v \in R}\Big(1 - \frac{q\varepsilon}{v}\Big). \tag{$*$}$$

Step 4: the normaliser does the same. Adjoining $u = \varepsilon$ to $\mathcal{E}^C$ adds, to the numerator, $\prod_{v\in R}(1-qv/\varepsilon)(1-q\varepsilon/v)\cdot(1-q)$ from $\prod_{j,k}(1-qu_k/u_j)$ and $\prod_k(1-q\varepsilon w_k)$ from $\prod_{j,k}(1-qu_jw_k)$; and, to the denominator, $\prod_{v\in R}(1-q\varepsilon v)$ from $\prod_{j<k}(1-qu_ju_k)$ together with one factor $(1-q)$ from $(1-q)^{n-r}$. The two $(1-q)$'s cancel, and $(1-qv/\varepsilon) = (1-q\varepsilon v)$ cancels against the $j<k$ product, leaving

$$\frac{\mathcal{E}^C_s}{\mathcal{E}^C_{s-1}}\Big|_{w_m=\varepsilon} = \prod_{k=1}^{N}(1-q\varepsilon w_k)\ \prod_{v\in R}\Big(1-\frac{q\varepsilon}{v}\Big),$$

which is $(*)$. Hence $\widetilde{A}_s/\widetilde{A}_{s-1} = 1$. □

Remark 10 (verification, with a plant). The two ratios of Steps 3 and 4 were checked *separately* against their closed forms — not merely their quotient, which could be 1 with both wrong — over all $(N, \mathbf{M}, m, \mathbf{I}, \varepsilon)$ with $N \le 3$ and $s-1 \ge r$: 112/112 for F, 112/112 for $\mathcal{E}^C$, 112/112 for $\widetilde{A}$ itself (`src_proof_check.py`). The proof uses $\varepsilon^2 = 1$ in three places, so the plant is $\varepsilon = 2$: the $\widetilde{A}$ equality must *fail* there, and it fails 24/24. A green at $\varepsilon = 2$ would have meant the instrument was blind.

With multiplier 1 the weight filtration collapses ($W_d = W_0$ for all d), so $\phi_\lambda\phi_\mu$ and the span sit at the same weight and the baseline constants have no short-root poles — which is what the baseline census shows. A renormalisation has short-root weight $\chi(\varepsilon)$:

$$\Psi_\chi(\mathbf{M})\big|_{w_m=\varepsilon} = \chi(\varepsilon)\,\Psi_\chi(\mathbf{M}^-)\big|_{w_m=\varepsilon},$$

so the condition for $\Psi_\chi\phi_\lambda\phi_\mu$ to stay in the short-root lattice is

$$\textbf{(SR)} \qquad \chi(\varepsilon) = 1 \quad \text{for every } \varepsilon \text{ with } \varepsilon^h = 1.$$

Proposition 11 (short-root lattice lemma)**.** $\mathrm{span}_{\mathcal{O}_{m,\varepsilon}}\{\phi_{\mathbf{I}}\} = \{f : (f(\mathbf{M}) - f(\mathbf{M}^-))|_{S^\varepsilon_m} = 0\ \forall \mathbf{M} \ni m\}$, *assuming* (SRC) = *Theorem 9. Consequently if* χ *satisfies* (SR) *then every* $\widehat{c}^{\,\nu}_{\lambda\mu}$ *is regular at* S^ε_m.

Proof. Verbatim the proof of Proposition 5 with g replaced by 1 and $\mathbf{M}''$ by $\mathbf{M}^-$. The two counts: the defect map onto $k^{\{\mathbf{M}\ni m\}}$ is surjective by the same indicator argument, giving colength 2^{N-1}; and $v_{S^\varepsilon_m}(\widetilde{A}_{\mathbf{I},\mathbf{I}}) = 1$ iff $m \in \mathbf{I}$ — the factor $(1-w_m^{-2})$, which is the $j=k$ term of (M4)'s $\prod_{j\le k}$ — and no other positive-root factor vanishes on $\{w_m = \varepsilon\}$ at generic q, since $(1-qw_j^2)$ needs $w_j^2 = q^{-1}$ and the A-type and q-pair factors cut out different divisors. So $v(\det) = \#\{\mathbf{I} \ni m\} = 2^{N-1}$. For the consequence: $f = \Psi_\chi\phi_\lambda\phi_\mu$ has $f(\mathbf{M})|_S = \chi(\varepsilon)\cdot 1\cdot f(\mathbf{M}^-)|_S = f(\mathbf{M}^-)|_S$. □

Remark 12 (inevitability). The two conditions are met by ψ *for the same reason.* From `eq:dict`, $\rho^X - 1 = q(t-1)^2/((1-qt)(1-q/t))$ with $t = w_a^h$, so $\{\rho^X = 1\} = \{q = 0\} \cup \{w_a^h = 1\}$: the obstruction vanishes exactly on the h-th roots of unity. And $\psi(v) = (1-qv^h)/(1-q)$ satisfies $\psi(\varepsilon) = 1$ exactly when $\varepsilon^h = 1$. **The locus where the multiplier is already trivial is precisely the locus where the renormaliser is required to be trivial.** The short-root divisors of $\det\widetilde{A}$ are $\{w_m^h = 1\}$ — the same set. I did not arrange this; I ran an arm that broke, and the diagnosis predicted that it must break at $\varepsilon = -1$ and *not* at $\varepsilon = +1$, which is what the data show.

8 Theorem C: the exact family

Theorem 13 (exact characterisation)**.** *Let* $\chi \in \mathbb{C}(q)(v)^\times$ *be regular and non-vanishing at the generic point of each divisor below,* $\Psi = \Psi_\chi$*, and put* $\kappa := \chi(x)\chi(1/x) \in k^\times_{ab}$*. Fix* λ, μ.

1. (long roots) *For $a < b$, the following are equivalent:*

   (a) *every $\widehat{c}^{\nu}_{\lambda\mu}$ is regular at H_{ab};*

   (b) *$(\kappa g - 1) \cdot (\phi_\lambda \phi_\mu)(\mathbf{M}'')|_H = 0$ for every $\mathbf{M}'' \subseteq [N] \setminus \{a, b\}$;*

   (c) *$\kappa = g^{-1}$, or no $\mathbf{M}'' \subseteq [N] \setminus \{a, b\}$ satisfies both $\mathbf{M}'' \geq \lambda$ and $\mathbf{M}'' \geq \mu$.*

2. (short roots) *For $\varepsilon^h = 1$, every $\widehat{c}^{\nu}_{\lambda\mu}$ is regular at S^{ε}_m if $\chi(\varepsilon) = 1$; and if $\chi(\varepsilon) \neq 1$ it is regular there only when $(\phi_\lambda \phi_\mu)(\mathbf{M}^-)|_S = 0$ for all $\mathbf{M} \ni m$.*

Hence the renormalisations that are pole-free on all *these divisors for* all *(λ, μ) are exactly*

$$\boxed{\chi = \psi \cdot \eta, \qquad \eta(x)\eta(1/x) = 1, \qquad \eta(\varepsilon) = 1 \ \ \forall\, \varepsilon^h = 1.}$$

Proof. (1) Put $f := \Psi \phi_\lambda \phi_\mu$. By (QC) and the computation in Theorem 7, $f(\mathbf{M})|_H = \kappa g^2 f(\mathbf{M}'')|_H$, so the defect is $D_{\mathbf{M}}(f) = (\kappa g^2 - g) f(\mathbf{M}'')|_H = g(\kappa g - 1) f(\mathbf{M}'')|_H$. By Proposition 5, (a) $\iff$ $D_{\mathbf{M}}(f) = 0$ for all $\mathbf{M}$, and $g \neq 0$ gives (b). For (b) $\iff$ (c): if $\kappa = g^{-1}$ then $\kappa g - 1 = 0$. Otherwise $\kappa g - 1 \neq 0$ in k_{ab}, and since $\Psi(\mathbf{M}'')|_H \neq 0$ by hypothesis we need $(\phi_\lambda \phi_\mu)(\mathbf{M}'')|_H = 0$ for all $\mathbf{M}''$. That holds iff no $\mathbf{M}'' \geq \lambda, \mu$: "$\Leftarrow$" is (M3); "$\Rightarrow$" is Rick's Lemma L (his §2bis: a degeneration $w_c = t^{-e_c}$, $t \to 0^+$, in which $\phi^0_\kappa(\mathbf{M}'') \to 1$ when $\mathbf{M}'' \geq \kappa$), which shows $\phi^0_\lambda \phi^0_\mu(\mathbf{M}'')|_H \neq 0$ at $q = 0$ and hence $\neq 0$ at generic q.

(2) is Theorem 9 and Proposition 11, with the same defect computation at $g = 1$.

For the final display: (COB) is $\kappa = g^{-1}$, and ψ is one solution by Theorem 2, so the solution set of (COB) is the coset $\psi \cdot \{\eta : \eta(x)\eta(1/x) = 1\}$; (SR) is $\chi(\varepsilon) = 1$, and $\psi(\varepsilon) = 1$ for $\varepsilon^h = 1$ (Remark 12), so it transfers to $\eta(\varepsilon) = 1$. □

Corollary 14 (the brief's Theorem C, answered negatively)**.** *The long-root pole order is* ***not*** *invariant across the family of diagonal renormalisations. By Proposition 5 it is at most* 1*, and by Theorem 13(1) it is* 0 *on the coset $\psi \cdot \{\eta\}$ and* 1 *off it (on every non-degenerate (λ, μ)). The brief asked for an invariance theorem that would bind Rick's unwritten geometric normalisation; instead the pole-removing normalisations are pinned down exactly, and any geometric normalisation that is diagonal in this sense* must *lie in the boxed coset.*

Remark 15 (monomial twists). $\eta(v) = v^k$ satisfies $\eta(x)\eta(1/x) = 1$ for every k, and $\eta(\varepsilon) = \varepsilon^k$, so in type C ($h = 2$, $\varepsilon = \pm 1$) the admissible monomial twists are exactly the *even* powers. This was used as a falsifiable prediction: $\chi = v\psi(v)$ must fail at $w = -1$ only, and $\chi = v^2\psi(v)$ must be clean. See §9.

9 Verification

All computations: pure SymPy, engine `proofs/code-1017c2-coboundary/q484_engine.py` (the 2026-10-17 copy, the only one carrying $\mathcal{E}^B$; for type C it is byte-identical in the relevant code path to the copies used on 10-15 and 10-16, checked by `diff`). Census: every (λ, μ, ν) with $\mu \geq \lambda$ in the linear extension and $c^{\nu}_{\lambda\mu} \neq 0$; $N = 3$, type C; 115 non-zero constants. A constant is counted *non-Laurent* iff, after `cancel`, its denominator has a non-monomial irreducible factor.

Instrument controls.

- **Baseline reproduces.** $\chi = 1$ at symbolic q returns 13/115 non-Laurent, which is the 2026-10-16 census number, and the poles are on long-root divisors only.

- **The q-specialisation is faithful.** $\chi = 1$ at $q = 3/7$ returns 13/115 *and the symmetric difference against the symbolic run is empty* — the same 13 triples with the same denominators, not merely the same count. This licenses reading the other arms at $q = 3/7$.
- **The instrument can see movement in both directions**: one arm removes poles (ψ), one arm adds them ($v\psi$). A green that could only ever be green is not evidence.

Arms. Each χ gives $\Psi_\chi(\mathbf{M}) = \prod_{m\in\mathbf{M}} \chi(w_m)$ and a full recomputation of all 115 constants from $\widehat{A}$ by the triangular solve — an independent route to $\widehat{c}$, not the $\ell_\nu(\Psi\phi_\lambda\phi_\mu)$ rewriting used in the proof. Predictions were written to `PREDICTIONS.md` before the arms finished.

arm	$\chi(v)$	(COB)	(SR)	result
`id`	1	fail	holds	13/115; long-root only
`psi`	$(1-qv^2)/(1-q)$	holds	holds	**0/115**
`x2psi`	$v^2(1-qv^2)/(1-q)$	holds	holds	**0/115**
`xpsi`	$v(1-qv^2)/(1-q)$	holds	fails at -1	57/115; **zero** long-root, all at w_i+1
`x`	v	fail	fails at -1	57/115; long-root *and* w_i+1
`sq`	$((1-qv^2)/(1-q))^2$	fail	holds	13/115; **the same 13, same denominators**
`h1`	$(1-qv)/(1-q)$	fail	fails at -1	57/115; long-root *and* w_i+1
`unrm`	$1-qv^2$	fail	fails at ±1	57/115; long-root, w_i+1 *and* w_i-1

Every pre-registered prediction holds, at divisor level. The table's "result" column is not a count: each entry records *which divisor classes* carry poles. Reading the two condition columns against the three divisor classes:

- **(COB) governs the long-root divisors and nothing else.** The four arms with (COB) failing (`id`, `x`, `sq`, `h1`, `unrm`) all have long-root poles; the three with (COB) holding (`psi`, `x2psi`, `xpsi`) have *none*, including `xpsi`, which is riddled with poles of the other kind.
- **(SR) governs the short-root divisors, and separately at each ε.** The arms with $\chi(-1) \neq 1$ (`xpsi`, `x`, `h1`, `unrm`) have poles at w_i+1 and no others have any; `unrm` is the *only* arm with $\chi(+1) \neq 1$ and the *only* arm, of eight, that produces a w_i-1 pole.
- **The positive prediction was made before it was measured.** $\chi = v^2\psi$ satisfies both conditions and was predicted clean at a value of χ not previously run; it returns 0/115.
- **The sharpest set prediction holds.** Theorem 13(1) makes the long-root pole locus depend on χ only through the predicate $\kappa g = 1$, which fails for both `id` and `sq`; so `sq` must return not merely 13 but *the same* 13 triples. It does, with identical denominators on every one.

A grading lesson, paid for in this session. `x` and `xpsi` both return 57, and their sets of 57 triples are **identical** — symmetric difference empty. Grading by count would have called them the same; grading by set, which is my standing rule (`two-criteria-can-fit-the-same-count-and-differ-as-sets`), would *also* have called them the same. They are not: on exactly the 13 baseline triples their denominators differ, `x` retaining the long-root factor that `xpsi` has shed, e.g. at $(\emptyset, \emptyset, \{1,2\})$

$$\texttt{x}:\ (w_2+1)(w_1+1)(w_1w_2-1) \qquad \text{versus} \qquad \texttt{xpsi}:\ (w_2+1)(w_1+1).$$

The whole of Theorem 7 lives in that difference. So: *a set is itself a projection. When two arms agree as sets, descend to the label each element carries* — here, the pole divisor, not the triple. One level below the count-to-set rule, and the first time it has mattered.

Generic q. The main positive arm was also run at symbolic q rather than $q = 3/7$: $\chi = \psi$ over $\mathbb{C}(q, w_1, w_2, w_3)$ returns **0/115** as well, so Theorem 7 is confirmed at generic q and not only at a specialisation. The baseline was likewise run at symbolic q (13/115). The other six arms were read at $q = 3/7$, licensed by the empty symmetric difference of the two baseline runs.

10 Where the pre-registered mechanism broke

The brief's Theorem-B argument reads $\prod_{m\in\mathbf{M}} \psi(w_m)$ as a *diagonal rescaling of the basis*, and concludes that a structure constant is multiplied by a ratio of ψ's — a unit at H_{ab}, hence unable to cancel a pole. That conclusion is correct for the operation it names, and the operation it names is not the one in the formula.

- **Label-indexed** rescaling: $\phi_{\mathbf{I}} \mapsto \chi_{\mathbf{I}}\phi_{\mathbf{I}}$ with $\chi_{\mathbf{I}} \in F$. Then $\widehat{c}^{\,\nu}_{\lambda\mu} = (\chi_\lambda\chi_\mu/\chi_\nu)\, c^{\nu}_{\lambda\mu}$, a unit multiple; poles cannot move. *True.*
- **Point-indexed** rescaling: $\phi_{\mathbf{I}} \mapsto \Psi\phi_{\mathbf{I}}$ with $\Psi \in \mathcal{A}$ a single function of the *point*. Then $\widehat{c}^{\,\nu}_{\lambda\mu} = \ell_\nu(\Psi\,\phi_\lambda\phi_\mu)$, which is not a unit multiple of $c^{\nu}_{\lambda\mu}$ at all: multiplication by a fixed element of the algebra reshuffles coordinates.

The product $\prod_{m\in\mathbf{M}}$ is indexed by $\mathbf{M}$, the point. And the decisive point is that **Theorem A requires the point-indexed reading**: the collapse $\mathbf{M} \mapsto \mathbf{M}''$ acts on the point, so a label-indexed $\chi_{\mathbf{I}}$ cancels out of the collapse ratio entirely and changes no multiplier. So the brief proved Theorem A under reading 2 and refuted Theorem B under reading 1, in adjacent paragraphs, with the same symbol.

> **Rule.** The index set of a rescaling decides whether it is a change of basis or multiplication by an algebra element. Before reasoning about what a rescaling does to structure constants, say out loud which set the product runs over — and check that the *other* theorems in the same document are using the same reading.

This is a relative of the one-coordinate-match family (`the-missing-coordinate-is-one-rule-in-nine-drawe`) but it is a different animal and I want the distinction recorded. There, two *objects* agreed on one coordinate and I inferred identity. Here there is one object and two *readings of my own notation*, and the brief's two theorems required *incompatible* readings: the document was internally inconsistent while each paragraph was locally correct. A consistency check across paragraphs of one's own brief would have caught it; a check of any single paragraph would not.

11 Consequences

Retraction 1. *Q491's criterion — $c^{\nu}_{\lambda\mu}$ Laurent $\iff$ $|\nu| - \max(|\lambda|, |\mu|) \le 1$, measured* 225/225 *with zero false positives and zero false negatives — is a property of the $\mathcal{E}^C$-normalised basis $\widetilde{A}$, not of the algebra $\mathcal{A}$. At $N = 3$ every non-Laurent constant has its poles on long-root divisors only, so under the ψ renormalisation all* 115 *become Laurent and the criterion becomes vacuous. The criterion remains true and useful* of *$\widetilde{A}$; what is withdrawn is any reading of it as intrinsic.*

Retraction 2. *The negative half of Q500 — "pole $\iff$ Pfaffian category", with the pole as an obstruction — is withdrawn in the form I stated it. The pole is not an invariant of the algebra; it is an invariant of a choice of normaliser, and the choice that removes it is explicitly available. The identity* $\rho^X - 1 = q\, w_a^h([K^X(w_a)]_0^1)^2$ *is unaffected: it is a true statement about the* cochain ρ.

What replaces them is better, and is the actual content of today:

> The obstruction is a **cohomology class**, and it is **zero**. The poles are removable *because* ρ splits as $\varphi(w_a)\varphi(w_b)$. Had it not split — had it depended on a third w_c, or failed to factor by variable — Theorem 13 says no point-diagonal renormalisation could have removed the poles, and the obstruction would have been real. The non-vanishing of $\rho - 1$ and the existence of the poles are both statements about a representing cochain. What I had been calling the obstruction was the cochain.

This also sharpens Rick's sentence. He wrote that my poles *measure the normalisation defect*. Strictly that identifies two objects — the defect of the *basis* ($\rho \neq 1$) and the poles of the *structure constants* — and my pre-registered objection to the identification was the right instinct applied to the wrong case: they are different objects, and here they happen to be controlled by the same cochain, so trivialising one trivialises the other. The correct statement is the implication, not the identity: *ρ a coboundary $\Rightarrow$ the structure-constant poles are removable*, with Theorem 13 supplying the converse.

12 Gaps and questions

Gap 1 (type B, and $N \geq 4$)**.** *Theorems 2–13 are stated for general h and use only (QC), (M3), (M4) and (SRC); the type B input (QC) is proved (`thm:rhoB`). But the* census *is type C, $N = 3$ only. Q502 (re-measure every type B number against $\mathcal{E}^B$) is untouched. At $N = 4$ Rick reports that the first-order route behaves differently (his (b) and (c) both fail first at $N = 4$); nothing in §6 is first-order, so I expect Theorem 7 to survive, but it is unmeasured.*

Gap 2 (other divisors)**.** $\det \widetilde{A}$ *also vanishes on the A-type divisors $\{w_i = w_j\}$ and on q-divisors $\{q \cdot monomial = 1\}$. No baseline constant has a pole there, and no arm produced one, but I have no theorem.*

New questions. **Q510** — *is $\widehat{A}$ the weight function of a model?* $\widehat{A}$ is a basis with honest GKM collapse at both long and short roots and a genuine multiplication rule with no poles on them. Rick proposes it should be Koncki's twisted motivic Chern class for $LG(n, 2n)$. The test is cheap and sharp: $\psi|_{q=0} = 1$, so $\widehat{A}|_{q=0} = \widetilde{A}|_{q=0}$, and the two can only differ in their q-deformation; compare $\widehat{A}$ against Koncki's table for $LG(2, 4)$ on the entry Rick predicts is the odd one out, $(\{1, 2\}, \emptyset)$. He states he has not checked Koncki's table; neither have I. Held at `agent-summary`; a first-hand read is the prerequisite.

Gate Q504 observed. No novelty sentence is made here about Q484, Q491 or Q492. Digalaki–Wheeler, *Structure Constants for Free-Fermion Six Vertex Model Symmetric Functions* (Mittag-Leffler, July 2026, in preparation, no arXiv ID) is held only at `agent-summary`, and their hypothesis is the free-fermion locus, which I conjecture is my $\rho = 1$. Theorems 2, 7 and 13 concern my own normaliser and the removability of my own poles; Retractions 1 and 2 are retractions of my own claims. Neither is a priority claim.

Acknowledgement

Lemma QC, Proposition 5 and Theorem 6 of `2026-10-17-q488-converse-order-q1-proofs.pdf` are Rick's. The lattice lemma above is his Proposition 5, re-derived here with its two numerical inputs checked independently against my engine; Theorem 13(1)(c) uses his Lemma L. The hunch that the poles are a normalisation artefact is his, it was flagged by him as a hunch, I pre-registered against it, and he was right.